Kodewala Academy – Java Backend Tests

Complete Question Bank (Test 2)

Test 2 – Java Backend Test # 2 | 40 questions

Q1. Which keyword allows JVM to call the main method?

A. static
B. void
C. public
D. main

Correct answer: C. public
Explanation: main must be public so the JVM, which is outside the class, can access it.

Q2. Why is the main method declared static?

A. To improve memory
B. To avoid object creation
C. To make code faster
D. To return values

Correct answer: B. To avoid object creation
Explanation: The JVM calls main before any object exists, so it must be static and callable via the class name.

Q3. In Java, file name and public class name:

A. Can be different
B. Must be same
C. Need not match
D. Depend on OS

Correct answer: B. Must be same
Explanation: A file containing a public class must be named exactly after that class (Test.java).

Q4. Which of the following is a valid Java identifier?

A. 123name
B. employee-id
C. total_sum
D. first name

Correct answer: C. total_sum
Explanation: Identifiers cannot start with a digit, contain hyphens or spaces. Underscores are allowed.

Q5. Java identifiers are:

A. Case-insensitive
B. Numeric only
C. Uppercase only
D. Case-sensitive

Correct answer: D. Case-sensitive
Explanation: name and Name are different identifiers.

Q6. What is the default value of a boolean variable?

A. true
B. false
C. 0
D. null

Correct answer: B. false
Explanation: Instance/static boolean fields default to false.

Q7. Which variable is declared inside a method?

A. Instance Variable
B. Static Variable
C. Local Variable
D. Member Variable

Correct answer: C. Local Variable
Explanation: Variables declared inside a method/block are local variables.

Q8. Which variable belongs to an object?

A. Instance Variable
B. Static Variable
C. Local Variable
D. Final Variable

Correct answer: A. Instance Variable
Explanation: Each object has its own copy of instance variables.

Q9. There is only one copy of a _____ variable for the entire class.

A. Member Variable
B. Static Variable
C. Local Variable
D. Final Variable

Correct answer: B. Static Variable
Explanation: Static variables belong to the class and are shared by all objects.

Q10. Which operator is used for assignment?

A. ==
B. =
C. !=
D. >=

Correct answer: B. =
Explanation: = assigns a value; == compares.

Q11. Which operator returns true only when both conditions are true?

A. ||
B. !
C. ==
D. &&

Correct answer: D. &&
Explanation: Logical AND (&&) is true only if both operands are true.

Q12. Which of the following is a benefit of methods?

A. Code duplication
B. Reduced readability
C. Reusability
D. Increased complexity

Correct answer: C. Reusability
Explanation: Methods let you write logic once and reuse it, improving readability.

Q13. What is the output?

```
boolean a = true;
boolean b = false;
System.out.println(a || b);
```

A. true
B. false
C. 1
D. Compilation Error

Correct answer: A. true
Explanation: true || false is true.

Q14. Which of the following is a non-primitive data type?

A. char
B. long
C. double
D. String

Correct answer: D. String
Explanation: String is a class (reference type). The others are primitives.

Q15. Which access modifier allows access only within the same class?

A. protected
B. private
C. default
D. prblic

Correct answer: B. private
Explanation: private members are visible only inside their own class.

Q16. What is the purpose of command-line arguments?

A. Compile code
B. Pass dynamic input at runtime
C. Store bytecode
D. Create objects

Correct answer: B. Pass dynamic input at runtime
Explanation: They let you pass values to main(String[] args) when launching the program.

Q17. What will happen if the public class name and file name are different?

A. Runtime Error
B. Program executes normally
C. Compile Time Error
D. JVM Exception

Correct answer: C. Compile Time Error
Explanation: javac rejects a public class whose name does not match the file name.

Q18. What is the output?

```
int x = 10;
System.out.println(x > 5 && x < 20);
```

A. false
B. true
C. 30
D. Compilation Error

Correct answer: B. true
Explanation: 10 > 5 and 10 < 20 are both true.

Q19. Which memory optimization issue can occur if instance variables are used where static variables should be used?

A. Stack Overflow
B. Deadlock
C. Data Corruption
D. Increased Memory Consumption

Correct answer: D. Increased Memory Consumption
Explanation: Every object gets its own copy of an instance variable, wasting memory for data that could be shared.

Q20. Which statement about local variables is TRUE?

A. Can have access modifiers
B. Receive default values
C. Must be initialized before use
D. Exist throughout program execution

Correct answer: C. Must be initialized before use
Explanation: Local variables get no default value; the compiler requires initialization before use.

Q21. What is printed?

```
public static void main(String[] args)
{
    System.out.println(args.length);
}

// run as: java Test A B C
```

A. 2
B. 3
C. 4
D. Compilation Error

Correct answer: B. 3
Explanation: Three arguments (A, B, C) are passed, so args.length is 3.

Q22. Which of the following can be accessed without creating an object?

A. Instance Variable
B. Instance Method
C. Static Variable
D. Constructor

Correct answer: C. Static Variable
Explanation: Static members are accessed through the class name.

Q23. Which statement regarding protected members is correct?

A. Accessible only inside class
B. Accessible only inside package
C. Accessible everywhere
D. Accessible in same package and subclasses

Correct answer: D. Accessible in same package and subclasses
Explanation: protected = package access plus access from subclasses (even in other packages).

Q24. What is the output?

```
boolean a = false;
boolean b = true;
System.out.println(!(a || b));
```

A. 1
B. true
C. false
D. true and false

Correct answer: C. false
Explanation: a || b is true, and !true is false.

Q25. Which of the following is NOT a valid method name according to conventions?

A. calculateSalary()
B. printMessage()
C. sendEmail()
D. EmployeeData()

Correct answer: D. EmployeeData()
Explanation: Methods use camelCase starting lowercase. A leading capital is the convention for class names.

Q26. Which scenario is best suited for a static variable?

A. Employee Name
B. Account Balance
C. Shopping Cart Items
D. Company Name Shared by All Employees

Correct answer: D. Company Name Shared by All Employees
Explanation: Static fits data common to all objects; the others are per-object.

Q27. What is the primary purpose of Access Modifiers in Java?

A. To improve program execution speed
B. To control the visibility and accessibility of classes, methods, and variables
C. To reduce memory consumption
D. To convert source code into bytecode

Correct answer: B. To control the visibility and accessibility of classes, methods, and variables
Explanation: Access modifiers (public, protected, default, private) enforce encapsulation.

Q28. A banking application has a variable balance. The company policy states that balance should not be modified directly by any class outside BankAccount. Which access modifier should be used for the balance variable?

A. public
B. private
C. protected
D. default

Correct answer: B. private
Explanation: private restricts access to the declaring class; expose it via getters/setters.

Q29. An e-commerce application has these requirements: every product has its own price; all products share the same GST_RATE = 18%. Which design is most appropriate?

A. price → static, GST_RATE → instance
B. price → instance, GST_RATE → static
C. Both should be instance variables
D. Both should be static variables

Correct answer: B. price → instance, GST_RATE → static
Explanation: Price is per object (instance); GST rate is common to all (static).

Q30. An application stores employee IDs. The expected number of employees may grow to several million in the future. Which data type is the safer choice?

A. byte
B. boolean
C. short
D. int

Correct answer: D. int
Explanation: byte (±127) and short (±32,767) overflow; int handles about 2.1 billion.

Q31. A file-processing application needs the file path to be supplied at runtime because it changes between environments (Dev, QA, Prod). What is the best approach?

A. Hardcode the file path
B. Store it in a local variable
C. Take it as a command-line argument
D. Store it in a method

Correct answer: C. Take it as a command-line argument
Explanation: Command-line arguments let you supply environment-specific values without changing code.

Q32. A banking application performs this validation: if(balance > 0 && accountActive). If balance > 0 is false, what happens?

A. Both conditions are evaluated
B. Only second condition is evaluated
C. Compile-time error occurs
D. Second condition is skipped

Correct answer: D. Second condition is skipped
Explanation: && short-circuits: if the left side is false, the right side is not evaluated.

Q33. Which Eclipse shortcut is used to execute the current line of code and move to the next line during debugging?

A. F5
B. F6
C. F7
D. F8

Correct answer: B. F6
Explanation: F6 = Step Over.

Q34. During debugging in Eclipse, you are stopped at a method call and want to enter inside that method to debug its code. Which shortcut should you use?

A. F5
B. F6
C. F7
D. F8

Correct answer: A. F5
Explanation: F5 = Step Into.

Q35. During debugging in Eclipse, you want to continue program execution until the next breakpoint is encountered. Which shortcut should you use?

A. F5
B. F6
C. F7
D. F8

Correct answer: D. F8
Explanation: F8 = Resume.

Q36. While debugging in Eclipse, you have stepped into a method and now want to return to the calling method. Which shortcut should you use?

A. F5
B. F6
C. F7
D. F8

Correct answer: C. F7
Explanation: F7 = Step Return (finish the current method and return to the caller).

Q37. What will be the output?

```
class Student {
    int id;
    String name;
}
public class Test {
    public static void main(String[] args) {
        Student s = new Student();
        System.out.println(s.id);
        System.out.println(s.name);
    }
}
```

A. 0 / null
B. null / 0
C. Compilation Error
D. Garbage Value / Garbage Value

Correct answer: C. Compilation Error

Explanation: The quiz key marks Compilation Error as correct. By real Java behaviour the output would be option A (0 then null).

Note: Options A, B and D are two-line outputs (first line printed / second line printed). The quiz key looks wrong here. This code compiles and prints 0 then null (int fields default to 0, references to null), which was the option the student picked and was scored incorrect. Keep this in mind when revising.

Q38. What will happen?

```
class Employee {
    Employee(int id) {
        System.out.println(id);
    }
    public static void main(String[] args) {
        Employee e = new Employee();
    }
}
```

A. Default constructor will be automatically created
B. Object will be created with id = 0
C. Compilation error
D. Runtime exception

Correct answer: C. Compilation error

Explanation: Once any constructor is declared, the compiler no longer supplies the default one, so new Employee() has no matching constructor.

Q39. Which is the superclass of all classes in Java?

A. Class
B. Object
C. Main
D. System

Correct answer: B. Object

Explanation: java.lang.Object is the root of the class hierarchy.

Q40. What will happen?

```
// File 1
package com.a;
public class Test {
    int x = 10;
    protected int y = 20;
    public int z = 30;
}

// File 2
package com.b;
import com.a.Test;
public class Main {
    public static void main(String[] args) {
        Test t = new Test();
        System.out.println(t.x);
        System.out.println(t.y);
        System.out.println(t.z);
    }
}
```

A. Prints 10 20 30
B. Runtime Exception
C. Compilation
D. Prints 30 only

Correct answer: C. Compilation

Explanation: In a different package, default x is inaccessible and protected y is not accessible via a plain object (Main is not a subclass), so compilation fails. Only z (public) is accessible.